

DHANANJAYA SAMANTA SINGHAR

Software Engineer III

SDE-III @ JPMorgan Chase | 9 Years | Senior Backend & Platform Engineer | Distributed Systems • Microservices • Event-Driven Architecture • System Design | Java • Spring Boot • Kafka • AWS • Kubernetes • Data Engineering • AI Agents • RAG

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PROFESSIONAL SUMMARY

Experienced **Backend, Platform & Data Engineering** Professional with **9+ years** of experience, building scalable distributed systems, cloud-native applications, and event-driven platforms. Expertise in **Java, Spring Boot, WebFlux, Kafka, AWS, Docker, Kubernetes, Python, Spark,** Apache Iceberg, and Snowflake, with strong ownership of backend architecture, platform engineering, data pipelines, and cloud-native solutions. Proven experience in system design, performance optimization, scalability, resiliency, observability, and enterprise data platforms, with exposure to **GenAI, LLMs, RAG, LangChain, AI Agents, and MCP**. Currently working as **AVP / Software Engineer III at JPMorgan Chase & Co.**, contributing to Asset & Wealth Management (AWM) Technology – Data Engineering & Analytics.

TECHNICAL EXPERTIES

- **Backend & Platform Engineering -** Experienced in building scalable distributed systems, microservices, and cloud-native applications using Java, Spring Boot, WebFlux, Kafka, AWS, Docker, and Kubernetes, with a focus on performance, reliability, and scalability.
- **System Design & Architecture -** Strong experience in end-to-end system design, API architecture, event-driven systems, asynchronous processing, resiliency, observability, and production engineering, with ownership from design to deployment.
- **Cloud & Platform Engineering -** Experienced in developing and modernizing cloud-native platforms using AWS, Kubernetes, Docker, Terraform, and CI/CD, with a focus on automation and operational reliability.
- **API & Integration Engineering -** Experienced in designing and developing REST APIs, GraphQL services, distributed integrations, and event-driven communication for enterprise applications.
- **Security & Application Engineering -** Hands-on experience with OAuth2, JWT, Spring Security, HTTPS, SSL/TLS, and secure microservice communication.
- **Performance & Reliability Engineering -** Experience with performance tuning, concurrency, fault tolerance, retry mechanisms, asynchronous processing, scalability, and high-availability systems.
- **Observability & Production Engineering -** Experience with logging, monitoring, metrics, tracing, and production troubleshooting using Grafana, Kibana, Dynatrace, Datadog, Splunk, and CloudWatch.
- **Data Engineering & Distributed Data Systems** - Experience building data-intensive backend services, ETL pipelines, and distributed data platforms using Python, PySpark, Spark, Apache Iceberg, Snowflake, and SQL.
- **GenAI & AI-Driven Engineering -** Building solutions using LLMs, RAG, AI Agents, LangChain, Vector Databases, MCP, and Agentic AI for automation and developer productivity.
- **Technical Leadership & Project Ownership -** Demonstrated ability to handle requirements, stakeholder communication, architecture decisions, and end-to-end project delivery, taking ownership from solution design through production.
- **Agile & DevSecOps –** Experienced in working in **Agile development environments** and integrating **DevSecOps practices** across the software development lifecycle, with a focus on **CI/CD automation, security, code quality, and reliable production delivery**.

WORK EXPERIENCE

- Currently working in **J.P Morgan Chase & Co**, Bangalore as **SDE-III** from **Jul-2024.**
- Formerly worked in **A.P. Moller MAERSK** as a **Software Engineer JL3** from Oct-2021 to Jul-2024.
- Formerly worked in **Lowe's India, ITC Infotech & Allianz Technology**.

PROJECT DETAILS

J.P Morgan Chase & Co. *Software Engineer III*

07/2024 – Present *Bangalore, India*

Key Contributions:

- **Microservices & Event-Driven Architecture:** Designed and developed multiple high-performance, scalable Spring Boot microservices**,** including REST and GraphQL APIs, along with Kafka-based producers and consumers, enabling asynchronous processing and near real-time event propagation across downstream systems.
- **Technical Ownership:** Took technical ownership of backend platform components, contributing to architecture and platform design decisions for scalable and resilient distributed systems.
- **Performance Optimization:** Improved application performance and throughput by 10x (~200K to 3–4M events/hour) through concurrency optimization, JVM tuning, and monitoring.
- **Data Engineering:** Designed and built an EMR Serverless-based data snapshot platform that supports dynamic ingestion of SQL and NoSQL data**,** and developed PySpark-based ETL jobs on AWS EMR to process and load large volumes of data into Snowflake**.**
- **Production Support & Troubleshooting:** Worked on production deployments, performance monitoring, debugging application issues, and resolving production incidents, ensuring system stability and smooth operation.
- **GenAI:** Built GenAI-powered GraphQL agents and synthetic data generation solutions using Python, LLMs, RAG, LangChain, and prompt engineering.

A.P. Moller - Maersk. *Software Engineer JL3*

10/2021 - 07/2024 *Bangalore, India*

Key Contributions:

- **Microservices & Event-Driven Architecture:** Designed and developed Spring Boot WebFlux microservices using reactive and event-driven architecture, integrating Apache Kafka with Avro, SSL/TLS, retry and Dead Letter Topic (DLT) mechanisms, and Temporal workflows for reliable and reentrant processing across distributed applications.
- **Platform Engineering:** Developed reusable Security and Exception Handling libraries to eliminate boilerplate across microservices, while maintaining high code quality.
- **Security & Cloud-Native Engineering:** Ensured production-ready software through TDD using JUnit, BDD automation using Cucumber, along with Sonar remediation and Trivy security vulnerability fixes, deployed and managed containerized applications using Kubernetes.

Lowe's, India *Software Engineer*

03/2020 - 10/2021 *Bangalore, India*

Key Contributions:

- **Microservices & Event-Driven Architecture:** Developed reactive, non-blocking REST APIs using Spring WebFlux and integrated third-party REST services using WebClient, implemented Apache Kafka and GCP Pub/Sub for event-driven audit data processing and integrated multi-bucket Couchbase connectivity with Spring Boot.
- **Identity & Security Platform:** Worked with ForgeRock as an OAuth authorization server for JWT-based authentication, configured SAN certificates for ForgeRock AM and DS components, and contributed to securing and modernizing the Identity Platform.
- **Cloud, Performance & Quality Engineering:** Deployed containerized applications to Google Cloud Platform (GKE) using Spinnaker pipelines, performed application performance testing using Apache JMeter, and contributed to Sonar remediation, unit testing, and integration testing.

AWARDS & RECOGNITIONS

Business Impact Award (12/2025)

- *Recognition of Outstanding Contribution to AWM Tech India, J.P Morgan Chase & Co.*

Hackathon Outstanding Performer (11/2025)

- *Recognized for demonstrating proactive and exceptional effort in the GraphQL Ref Data GPT AI Agent with FAISS (Vector Indices) hackathon at J.P Morgan Chase & Co.*

Star Award (10/2022)

- *Recognized for exceptional performance and valuable contributions to the OHM Project at A.P. Moller - Maersk*

Valuable Player Award (12/2017)

- *Recognized as a key contributor to the Allianz Turkey Project.*

EDUCATION

Computer Science & Engineering (B. Tech-2015)

Biju Patnaik University of Technology, Rourkela, Odisha, India.